☼ Set Operations :

Union of two sets : The union of two sets A and B is the set consisting of elements which belong to A or B or to both and is written symbolically as $A \cup B$.

$$\therefore \quad A \cup B = \{x / x \in A \text{ or } x \in B\}.$$

Intersection of two sets : The intersection of two sets A and B is the set consisting of element which belong to A and B both and is written symbolically as $A \cap B$.

$$\therefore \quad A \cap B = \{x / x \in A \text{ and } x \in B\}.$$

☼ Disjoint sets :

Two sets A and B are said to be disjoint or mutually exclusive if and only if there is no element common to A and B.

i.e. $A \cap B = \phi$.

☼ Equality of sets or Equal sets :

Two sets A and B are said to be equal iff every element of A is an element of B and also every element of B is an element of A.

i.e. both of them have the same elements and symbolically we write $A = B$.

☼ Equivalent sets :

Two sets are said to be equivalent if they contain the same number of elements.

Note : Equal sets are always equivalent but equivalent sets are not always equal.

☼ Difference set :

Let A and B be two sets. Then the set of all elements which are in A but not in B is called the difference set of A and B and is denoted by $A - B$. Thus, $A - B = \{x / x \in A \text{ and } x \notin B\}$.

Note : (i) $A = B \Rightarrow A - B = \phi$

(ii) $A - B = A \cap B'$

(iii) $A - \phi = A$

(iv) $(A - B) \subset A$ and $(B - A) \subset B$

(v) $A \subset B \Rightarrow A - B = \phi$

(vi) $U - A = A'$.

☼ Symmetric Difference set :

Let A and B be two sets. The Set of all elements which are in A or in B but not in both is called the symmetric difference set of A and B and it is denoted by $A \Delta B$.

Thus, every member of $A \Delta B$ is either in A or in B that is in $A \cup B$ but not in both, so not in $A \cap B$.

$\therefore \quad A \Delta B = (A \cup B) - (A \cap B)$.

☼ Ordered pair :

Let A and B are two given sets. If we take an element $a \in A$ and $b \in B$ and write (a, b), the first element taken from A and the second from B, then (a, b) is called an ordered pair.

☼ Cartesian product of sets :

If A and B are two non - empty sets, then the set $\{(a, b) / a \in A, b \in B\}$ is called the cartesian product of A and B and is denoted by $A \times B$ (Read as 'A cross B').

$$\therefore \quad A \times B = \{(a, b) / a \in A, b \in B\}.$$